

Indian Institute of Science Education and Research Mohali

Department of Physics

Computational Analysis of Optical Spectroscopic Data of Active Galactic Nuclei Using FITS Files

Summer Project Report

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Abstract

This project report outlines the implementation of modern data reduction workflow in high energy astrophysics, following a dual computational approach and theoretical modeling. On the computation side, we have read and learn about the Flexible Image Transport System (FITS) files-the standard form of exchanging observational astronomical data between different institutions irrespective of the software and hardware environments. Utilizing Python and its appropriate libraries :- `astropy.io.fits`, `numpy`, `matplotlib.pyplot`, and `pandas`, we have extracted, represented, and analyzed plots of multi wavelength observational datasets. This data pipeline is validated by accessing the Fermi Gamma-ray Space Telescope archival database to reproduce blazar light curves and spectral emission distributions (SEDs). On the analytical side, we explore fundamental principles of high energy spectroscopy, focused on identifying continuous radiation profiles and atomic emission lines. By mapping these spectral features, we probe the physical conditions. This combined framework establishes the precise methodology required to handle, reduce, and interpret uncompressed archival data from diverse space-based observatories.

1 Chapter 1: Introduction

1.1 Blazars



Figure 1: Schematic representation of a blazar featuring a central supermassive black hole, surrounding dusty torus accretion material and bipolar, relativistic plasma jets.

Blazars are highly compressed, extremely bright Active Galactic Nuclei (AGNs) which shoot high energy jets of matter directly towards the Earth. They are powered by a massive central black hole which continuously engulfs surrounding matter like the gas, dust and stars. This matter then swirls into a rapidly spinning accretion disk, this disk radiates massive amount of energy into the surroundings. The region has powerful magnetic fields which guide the charged particles into emerging out of the poles as relativistic jets which travel at nearly the speed of light.

1.1.1 Spectroscopic classification of Blazars

Blazars are classified broadly into two subclasses based on observational features of their optical spectra

- 1. Flat Spectrum Radio Quasars (FSRQs):** These sources are characterized by the presence of prominent, strong, and broad emission lines superimposed on the non-thermal continuum. They are surrounded by enormous high energy gas clouds which create a "Broad Line Region" cloud network.
- 2. BL Lacertae (BL Lac) objects:** These sources can be defined as Active Galactic Nuclei with a relativistic jet pointed directly towards the Earth and are distinguished specifically by a featureless optical spectrum that lacks strong emission or absorption lines.

1.1.2 Why are Blazars chosen for this project?

Blazars are chosen as the primary targets of observation for this project because they are highly variable, multi wavelength emitters and are one of the brightest objects in the observable universe. Because they dominate the high-energy gamma ray sky observed by the Fermi Gamma-ray Space Telescope we can have rich mountain of live data to analyze.

1.2 Principles of Spectroscopy

Spectroscopy is the science of splitting white light into its component lights to read the information car-

ried by it. By measuring the brightness of each individual wavelength hitting a telescope detector, astronomers can figure out exactly what a cosmic object is made up of, what its temperature is and how fast it is moving.

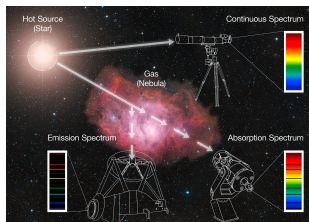


Figure 2: Basic principle of astronomical spectroscopy. Light from an astronomical object is dispersed into its constituent wavelengths, producing a spectrum that reveals information about the object's composition, temperature and motion

1.2.1 The Continuum (The Background Light)

The continuum is a smooth, unbroken smear of background light across all wavelengths. It acts as the ground baseline of our graph. In a blazar, this smooth light isn't caused by heat like in a regular star. It is caused by the relativistic jets of charged particles moving in circles in near speed of light by strong magnetic fields. Because the magnetic fields control the background continuum light, it becomes highly organized. The **Stokes Q** and **Stokes U** graphs are physically tracking the exact alignment of vibration of this background continuum light.

1.2.2 Emission lines:

The emission lines are the sharp narrow spikes that rise straight out of the smooth background continuum graph. Atoms in the gas clouds have electrons orbiting them all the time. When energy hits these electrons they get excited and jump to higher orbits. When they de-excite and return to their original orbits, they release a packet of light with a very specific wavelength. Because each element has its own electronic structure, each element shoots out spikes

at completely different and predictable wavelengths. This acts like the chemical fingerprint in space and helps us identify the particular element just by looking at the graph.

1.2.3 The Doppler Effect:

In space the gas clouds never stay still, they are always orbiting the supermassive black holes in speeds of millions of kilometers per hour. Due to the Doppler effect the light coming from the gas coming towards us gets squished into relatively shorter wavelengths, while the light coming from gas moving away from us gets stretched to relatively higher wavelengths. Because we are looking at billions of atoms which have a distribution of velocities with different parts of the gas moving towards and away from us at different speeds simultaneously, the sharp needle like spikes get blurred into wide, flat bell-curves. Measuring the width of this curve gives us the ability to predict how fast the gas is rotating around the black hole.

2 Chapter 2: Computational Methodology

2.1 FITS

2.1.1 What is FITS file?

Flexible Image Transport System (FITS) is a standard format for exchanging astronomical data between institutions, independent of the hardware platform and software environment. A data file in FITS format consists of a series of Header Data Units (HDUs), each containing two components: an ASCII text header and the binary data. The header contains a series of header keywords that describe the data in a particular HDU and the data component immediately follows the header.

2.1.2 Header and data units

The first header in a FITS file is known as the primary header, and any number of extensions can follow the primary HDU. The data unit following the primary header must contain either an image or no

data at all, but each extension can contain one of several different data types, including images, binary tables, and ASCII text tables. The value of the XTENSION keyword in the extension's header identifies the type of data the extension contains.

2.1.3 Why astronomers use FITS?

Astronomers use FITS file format as the standard format because it is specifically designed to store not just pictures but also complex scientific data with detailed information about how the data was captured. FITS files use a Header + Data Unit (HDU) structure. The header is in plain text and contains crucial metadata—such as the telescope used, exposure time, celestial coordinates, and weather conditions. This ensures the raw data is never separated from its context. Unlike JPEGs or PNGs, FITS files can store multi-dimensional arrays, 3D data cubes, and complex data tables. This allows them to hold spectral data, radio wave frequencies, and volumetric measurements in a single file. Standard image formats compress data or convert it to standard 8-bit color, which destroys scientific value. FITS preserves the raw, uncompressed measurements (like the exact number of photons or electrons captured by a pixel) for precise mathematical analysis. FITS has a strict "Once FITS, always FITS" rule. The International Astronomical Union (IAU) ensures that all updates to the format are backward-compatible so that decades-old telescope data can still be read by modern software.

2.2 Core Design Idea

The objective of this project is to analyze astronomical spectroscopy data stored in FITS files. The methodology involves reading the FITS file, extracting the header and spectral data, visualizing the spectrum, and interpreting the observed spectral features.

2.2.1 Algorithm Overview

- Reading the FITS file
- Extract the header information
- Retrieve the spectral data

- Plot the spectrum
- Analyze the spectral structure

2.2.2 Flow Diagram

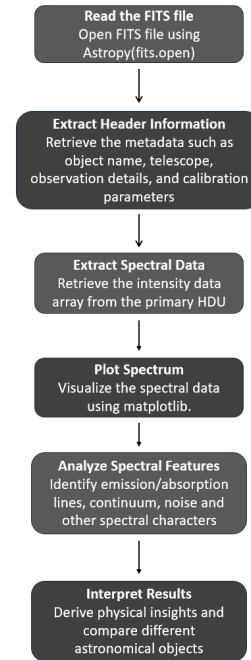


Figure 3: Representation of FITS flow

2.2.3 Key Code Logic

- Reading the FITS file

Listing 1: Opening a FITS file using Astropy

```

1 from astropy.io import fits
2 hdul = fits.open("mrk501.q.fits")
  
```

- Extracting the data

Listing 2: Retrieving the spectral intensity values from the primary HDU

```

1 data = hdul[0].data
  
```

- Plotting the Spectrum

Listing 3: Generating the spectrum for visual analysis

```
1 plt.plot(data)
```

2.2.4 Processing Workflow

```
1 BEGIN
2 For each FITS file
3 Open FITS file
4 Read Primary HDU
5 Extract header information
6 Extract spectral data array
7 Plot the spectrum
8 Analyze spectral features
9 Interpret results
10 END
```

3 Chapter 3: Results and analysis

3.0.1 PKS 0454-234

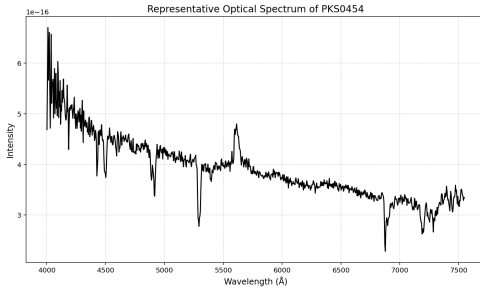


Figure 4: Representative Spectrum

The representative optical spectrum of PKS 0454-234 was obtained by extracting the spectral data from the FITS file and converting pixel positions into wavelength by calibrating using the information in the FITS header.

The spectrum covers the wavelength range of approximately 4000-7500 and shows a continuous variation of intensity with wavelength. Several prominent features can be observed from the overall spectral profile. This indicates variations in intensity at specific wavelength regions.

The extracted spectrum demonstrates the successful conversion of raw FITS spectral data into a wavelength vs intensity optical spectrum. This helps us to further analyze the spectral characteristics of the source.

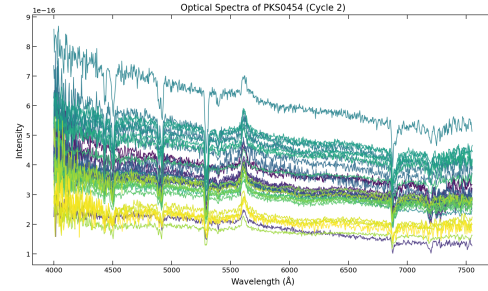


Figure 5: Optical Spectra of PKS 0454-234

The optical spectra of PKS 0454-234 from different observation epochs were compared to study changes in the observed intensity over time.

The spectra show variation in their overall intensity levels between different observations, indicating changes in the brightness of the source during the observation period. However, the general spectral shape remains similar across different epochs.

The consistency of the observed spectral features suggests that the wavelength calibration and extraction process produce reliable results, while the intensity variations indicate changes in the observed emission from the source.

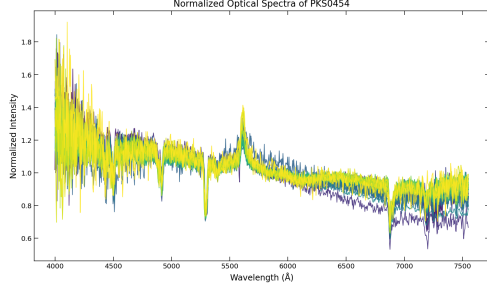


Figure 6: Normalized Spectra of PKS 0454-234

The normalized spectra were generated to compare the relative spectral shape of PKS 0454-234 independent of differences in overall intensity.

After normalization, the spectra show a similar profile across different observation epochs. This indicates that although the observed intensity changes, the relative distribution of intensity with wavelength remains largely consistent.

The normalized comparison helps distinguish between changes in overall brightness and changes in the spectral structure of the source.

3.0.2 PKS 0502+049

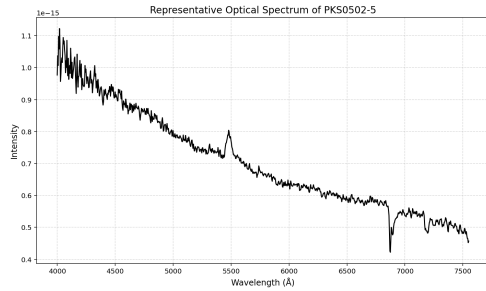


Figure 7: Representative Spectrum of PKS 0502+049

The representative optical spectrum of PKS 0502+049 was extracted from the FITS data using the same wavelength calibration procedure applied for PKS 0454-234.

The spectrum shows a continuous intensity distribution across the optical wavelength range with several variations at specific wavelength regions. The

extracted spectrum provides information about the observed spectral profile of PKS0502-5 and allows comparison with other observation epochs.

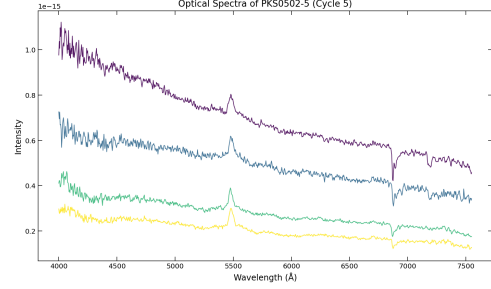


Figure 8: Optical Spectra of PKS 0502+049

The spectra obtained from multiple observation epochs show variations in intensity, indicating changes in the observed brightness of PKS 0502+049.

Although the flux levels differ between observations, the overall spectral structure remains similar. This suggests that the changes observed are mainly related to variations in intensity rather than major changes in the spectral profile.

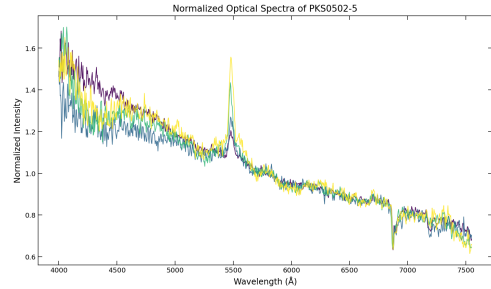


Figure 9: Normalized Spectra of PKS 0502+049

The normalized spectra of PKS 0502+049 show that the spectral profile remains relatively consistent after removing differences in overall intensity.

The similarity between normalized spectra indicates that the source maintains a comparable spectral structure across different observations, even when variations in brightness are present.

4 Chapter 4: Comparative Analysis

4.1 Comparison of Representative Spectra

The representative spectra of PKS 0454-234 and PKS 0502+049 were compared to examine their observed optical profiles. Both spectra span approximately the same wavelength range and show continuous intensity distributions with several noticeable features. PKS 0502+049 exhibits a higher overall intensity than PKS 0454-234 in the selected observations. Although both spectra display variations in intensity with wavelength, differences in flux levels and feature prominence are evident between the two objects.

4.2 Variability Analysis

The multi-epoch spectra of both objects show changes in intensity between different observations, indicating variability in their observed optical emission.

PKS 0502+049 shows comparatively larger differences in intensity between epochs than PKS 0454-234. However, the overall spectral profiles remain similar, suggesting that the variability mainly affects the intensity level rather than significantly changing the spectral structure.

4.3 Normalized Spectra Comparison

The normalized spectra of PKS 0454-234 and PKS 0502+049 were compared to examine their spectral profiles independent of intensity variations.

The normalized spectra show that both objects maintain relatively consistent spectral shapes across different observations. This indicates that the relative distribution of intensity with wavelength remains stable even when changes in overall brightness occur.

computational techniques based on FITS file processing. The spectral data were extracted, wavelength calibrated using FITS header information, and visualized through representative, multi-epoch, and normalized spectra.

The analysis successfully demonstrated the extraction and processing of astronomical spectral data. Both objects showed continuous optical spectral profiles with variations in intensity across different observations. Comparison of multi-epoch spectra revealed changes in observed brightness, while normalized spectra showed that the overall spectral structure remained relatively consistent.

This work demonstrates the application of computational methods for astronomical spectroscopy and provides a foundation for further analysis involving spectral line identification, redshift estimation, and detailed physical interpretation of astronomical sources.

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- [5] Source of the blazar image: <https://share.google/VGgzf6srchR4oH7yJ>
- [6] Source of the spectroscopy/workflow image: <https://share.google/Wi1BV39pvGfo7Z5wZ>

5 Chapter 5: Conclusion

In this project, optical spectroscopy data of PKS 0454-234 and PKS 0502+049 were analyzed using